

Forward Data Lab Application — Summary Report

Asta vs. Google Scholar Evaluation, Workflow Analysis, and Cluster-Aware Visualization. Condensed summary; see the full report for methodology, figures, and detailed discussion.

1. Asta vs. Google Scholar: Empirical Comparison

The evaluation compares Asta Find Papers (Allen AI) with Google Scholar on a controlled 50-query FinTech suite, built from 25 curated papers with two query phrasings each and embedded ground-truth metadata. Both engines are restricted to their Top-10 per query (500 Asta papers, 473 Google Scholar). Asta is measured along seven directly inspectable axes: five comparable across both engines and two Asta-only diagnostics.

Table 1. Asta vs. Google Scholar across seven metrics (Top-10 per query). The Asta-vs-GS column is a ratio with Google Scholar at 1.0; Currency is reported in years.

#	Metric	Asta	Google Scholar	Asta vs GS
1	Overlap	—	—	0.12 mean (20/50 zero overlap)
2	Topical Hit Rate	0.65	0.80	0.81
3	Topical Breadth (/8)	1.52	1.30	1.17
4	Currency, mean year	2022.25	2022.07	+0.18 years
5	Citation Influence, log(c+1)	3.08	2.93	1.05 (approx. tied)
6	Evidence Depth	Abstract 9% / other 91%	n/a	Asta-only capability
7a	Self-Confidence Calibration	0.66	n/a	Asta-only diagnostic
7b	Summary Concept Coverage	0.49	n/a	Asta-only diagnostic

Top-10 overlap is very low (mean 0.12; 40 percent of queries share no result), so the two engines deliver substantially different bibliographies rather than re-orderings of one set. Google Scholar leads on Topical Hit Rate (0.80 vs 0.65, $p < 0.001$), the expected advantage of lexical matching on keyword queries, though partly inflated by the keyword dictionary used for classification. Asta leads on Topical Breadth (1.52 vs 1.30 sub-areas, +17 percent, $p = 0.055$, marginal) and is essentially tied on Citation Influence ($p = 0.27$). Mean publication year is tied, but the band distribution diverges: Google Scholar is bimodal (37 percent 2025+, 17 percent pre-2020) while Asta concentrates in 2023 to 2024 (38 percent). Asta uniquely surfaces attributed evidence snippets, only 9 percent of which come from abstracts, indicating retrieval that reaches into paper bodies. Two Asta-only diagnostics (Self-Confidence Calibration 0.66, Summary Concept Coverage 0.49) indicate non-trivial noise in Asta’s LLM-generated layer, though both are lower bounds constrained by title-only classification.

Overall, Google Scholar is the stronger tool for lexically anchored known-item retrieval and recency monitoring, and Asta the stronger tool for multi-disciplinary exploration and evidence-grounded reading. The two are best treated as complementary; Asta’s LLM-generated labels and summaries should be verified against its surfaced snippets.

2. How Asta Find Papers Works

Asta is an AI research assistant from the Allen Institute for AI; its Find Papers mode runs Retrieval-Augmented Generation over the Semantic Scholar corpus through a multi-stage, manually-coded pipeline (AstaBench, Appendix F.3). The flow is Query Analyser, then Execution Planner, then a semantic sub-workflow (retrieval, citation expansion, relevance judgment), then Final Ranker, with a separate Generate Report mode. Table 2 maps each component to its closest academic reference and the key architectural difference.

Table 2. Asta Paper Finder components, closest academic references, and key architectural differences.

#	Component	Closest Academic Reference	Key Architectural Difference
0	Overall Paradigm (RAG)	[3] Lewis et al.: single-pass RAG	Asta extends base RAG with iterative citation expansion and structured query decomposition.
1	Query Analyser	[5] Gao et al. (HyDE): one hypothetical document	Asta decomposes the query into typed fields and multiple rephrasings, handling metadata constraints explicitly.
2	Execution Planner / Semantic Sub-Workflow	[1] Cohan et al.: SPECTER citation-graph embeddings	Asta adds multi-query parallelism, cited-paper expansion, and iterative snowballing on the SPECTER substrate.
3	Relevance Judgment	[6] Zheng et al.: LLM-as-a-Judge	Asta judges document relevance against structured sub-criteria; per-criterion decomposition is Asta-specific.
4	Final Ranker	[7] Sun et al.: RankGPT / [8] Yang et al.: Rank-K	Asta uses a deterministic formula over relevance tier, date, citation count, and modifiers; no LLM call described [2].
5	Evidence Extraction	[9] Gao et al.: ALCE retrieve-then-generate	Asta keeps a verbatim snippet and an LLM-generated summary as separate layers, not one citation-annotated text.
6	Report Generation (Scholar QA)	[10] Asai et al.: Self-RAG	Asta uses modular separate components; Self-RAG integrates retrieval decisions into the LLM via reflection tokens.

The 50-query suite empirically corroborates these component-level differences. Near-zero result overlap is consistent with the multi-rephrasing Query Analyser and citation-graph expansion; the Topical Breadth lift matches semantic expansion across citation neighborhoods; and the bimodal-versus-mid-band Currency profile is consistent with the SPECTER embedding substrate and the deterministic Final Ranker.

3. Cluster-Aware Visualization of Asta Paper Finder Results

Selected task. From the Asta workflow components, the selected task is Asta’s user-facing report and result presentation stage, where the retrieved, ranked, and relevance-judged paper set is delivered to the user, currently as a ranked list. Following the brief’s allowance to propose changes, the implementation adds a consumer-side layer without modifying Asta’s internal pipeline.

Why it matters. Report Generation is Asta’s only touchpoint with the user; it is exactly where Asta’s upstream strengths (per-paper rationale, tier judgements) must become usable. What the user ultimately receives and acts on is the final report.

Why it is challenging. Result sets vary from a handful of papers to over a hundred, and user intent varies (fact-check, deep read, or full survey). The 50 to 100 range is too many to read individually but too few to summarise without loss, and a linear ranking collapses multi-dimensional sub-topic structure onto a single axis.

State of the art. The paginated ranked list dominates academic-search result presentation (Google Scholar, Semantic Scholar, Asta itself); state-of-the-art work advances the ranking component, not the presentation format. The list does not naturally surface the richer LLM-generated signals Asta produces.

Solution. A Chrome MV3 extension captures Asta’s result payload and renders it as a radial cluster mind map. Concepts are extracted via TF-IDF, with title up-weighting, query-term removal, and universal-term filtering to handle homogeneous result sets; clusters come from k-means with k-means++ initialisation, and labels from c-TF-IDF with a minimum-presence floor. Classical methods suffice because Asta’s own per-criterion

summary text substitutes for embedding models, keeping the pipeline entirely in-browser. The highest-impact next step is LLM-based cluster naming.

References

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